

Yuan Tuo

Department of Economics
University of California, Davis

yuatuo@ucdavis.edu
<https://yua-tuo.github.io>
+1 5302312622

Education

2020–26	Ph.D., Economics, University of California, Davis (expected)
2018–19	B.Ec., Economics (First Class Honours), The University of Sydney
2017	Exchange, University of Wisconsin–Madison
2015–18	B.Ec., Economics, The Australian National University
2014–15	Wuhan University, China

Research Interest

Econometrics, Applied Econometrics

Working Papers

Optimal Treatment Under Spillover Effects and Noncompliance (Job Market Paper)

Abstract: This paper considers optimal treatment allocation in the presence of spillover effects in both compliance and outcomes. We show that an individual’s outcome depends on neighbors’ compliance types besides her own type. For optimal allocation, we either maximize welfare subject to a budget constraint or minimize cost subject to a welfare goal, based on the characteristics of the target population. We propose an optimal randomized treatment rule and show that it achieves uniform asymptotic welfare efficiency with an $n^{-1/2}$ convergence rate and is uniformly asymptotically feasible. We apply the proposed rule to an anti-conflict experiment in public middle schools in New Jersey and show that not incorporating both layers of spillovers can lead to misallocation of treatment.

Causal Inference in RCTs with Multiple Treatment Arms

Abstract: When there is more than one treatment group in an RCT with noncompliance, local average treatment effects (LATE) are not directly comparable because each treatment has different compliers. We provide a partial-identification method to compare two treatments in the presence of noncompliance. In particular, we derive bounds on the average difference between pairs of treatment effects for those who would comply with both treatments, as well as bounds for other subpopulations. We also apply our bounds to an experiment evaluating the impacts of different depression treatments on economic outcomes in India.

Work in Progress

Inference of Marginal Treatment Effect (MTE) Estimator (with Takuya Ura)

References

Takuya Ura

UC Davis
Associate professor
takura@ucdavis.edu

Shu Shen

UC Davis
Associate professor
shushen@ucdavis.edu

Colin Cameron

UC Davis
Professor
accameron@ucdavis.edu

Presentations

2025 California Econometrics Conference

2025 Midwest Econometrics Group Conference

Fellowships and Awards

2023 Dean's Graduate Summer Support Award, University of California, Davis

2021 First-Year Fellowship, University of California, Davis

Teaching Experience

2020–25 **Teaching Assistant, University of California, Davis**

Undergraduate

Analysis of Economic Data, Econometrics, Economic & Financial Forecasting,
Economics & Business Data Analytics, Principles of Microeconomics,
Principles of Macroeconomics, Intermediate Macro Theory, Public Finance

Graduate (Ph.D. level)

Econometric Methods, Cross-Sectional Econometrics

2019 **Tutor, The University of Sydney**

Introductory Microeconomics, Introductory Macroeconomics

Referee Service

Econometrics Review

Software Skills

R, Stata, Matlab

Languages

Mandarin (native), English (fluent), Japanese (JLPT N1)